

Probing binary supermassive black holes in quasars through spectropolarimetry

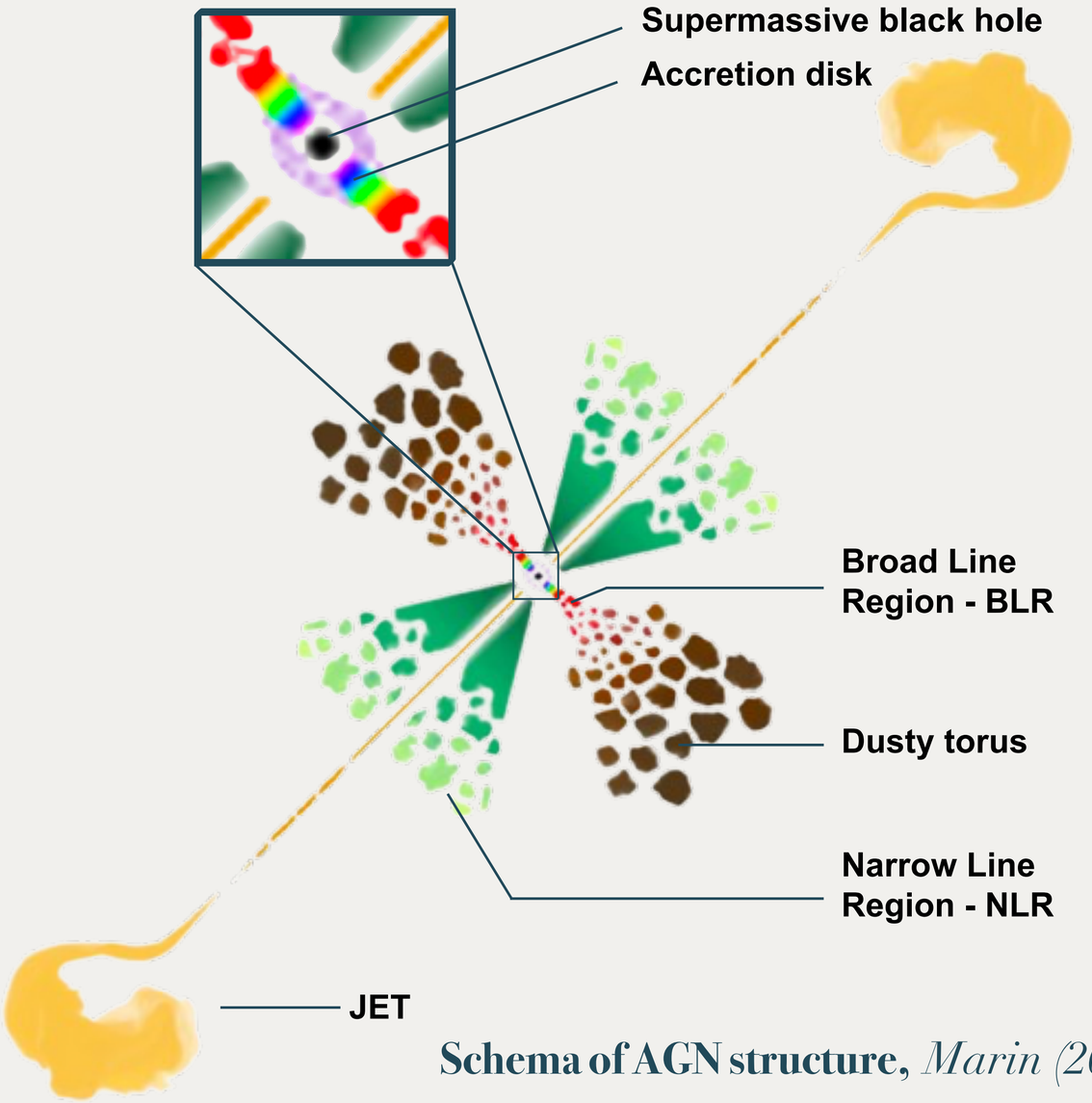
BIEDERMANN Julie and MARIN Frédéric
Astronomical Observatory of Strasbourg

SF2A - S09 : Studying galaxy evolution and cosmology in the context of future major observatories.
1 - 4 July 2025

Introduction - What is a quasar ?

Composition

- **Supermassive black hole + accretion disk**
  - $M_* = 10^6 - 10^{10} M_\odot$
  - Accretion disk responsible for the major part of the observed UV-near IR luminosity
- **Dusty Torus**
  - A donut-shaped structure that obscures parts of the quasar depending on the observer's line of sight.
- **BLR / NLR**
  - BLR $\Longrightarrow$ Interior of the torus
  - NLR $\Longrightarrow$ Exterior of the torus
- **Jets and Lobes**
  - Ionized matter at the speed of light



Schema of AGN structure, *Marin (2016)*

Introduction - Markarian 231 (Mrk 231)

Mrk 231

- **Ultra luminous infrared galaxy**
  - ➡ z = 0.042 (~ 175 Mpc)

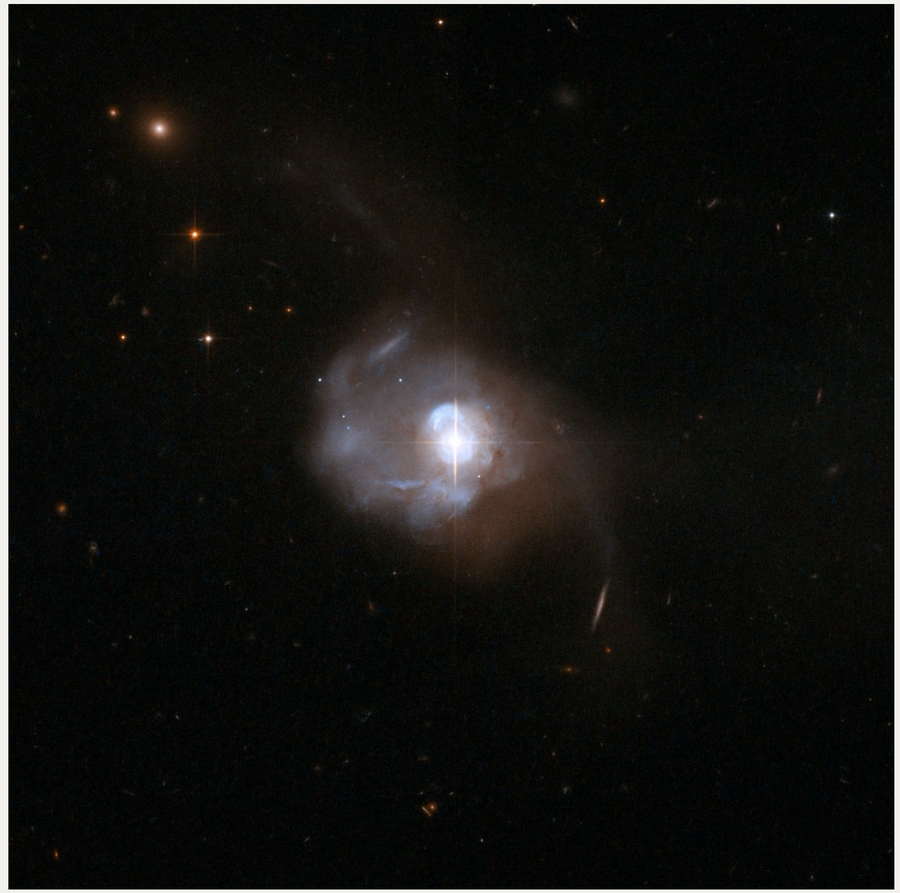
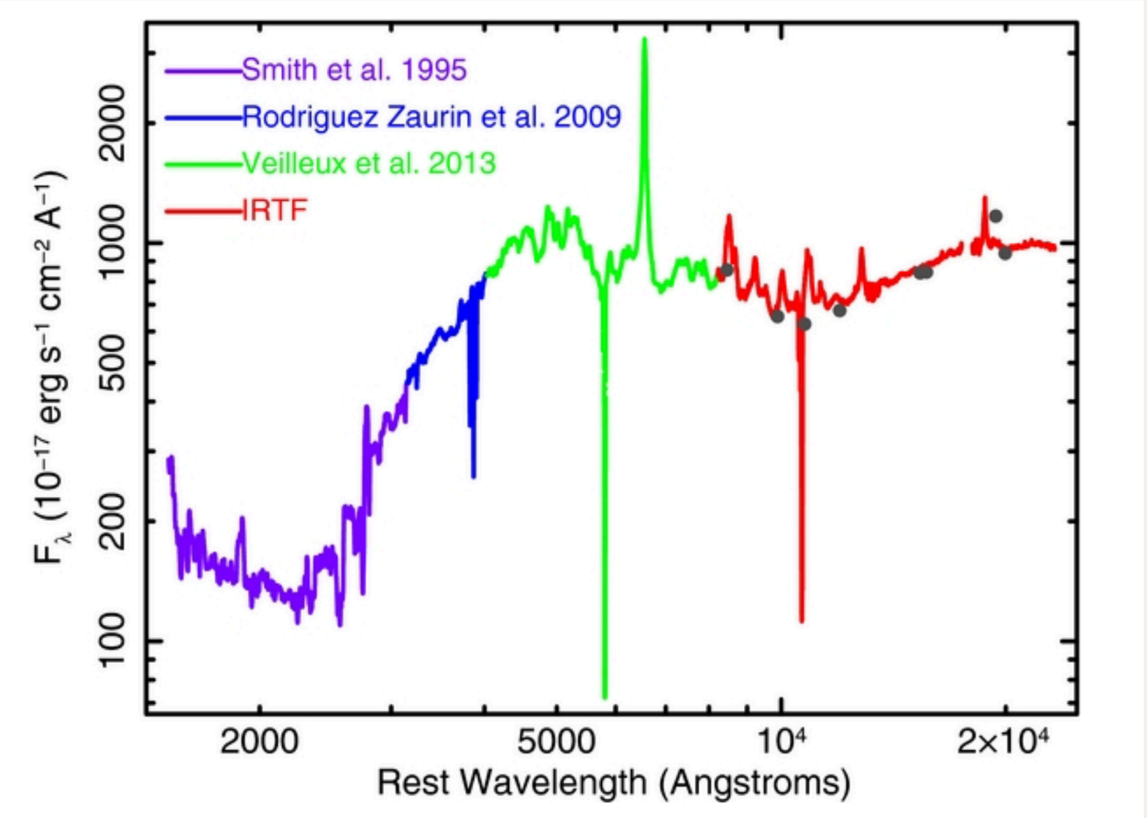


Image of Mrk 231, *Hubble Space Telescope*

Introduction - Markarian 231 (Mrk 231)

Mrk 231 - unique spectrum

- **Ultra luminous infrared galaxy**
  - ➡ z = 0.042 (~ 175 Mpc)
- **Unique UV-optical spectrum**
  - ➡ Cut off ~ 2400 Å

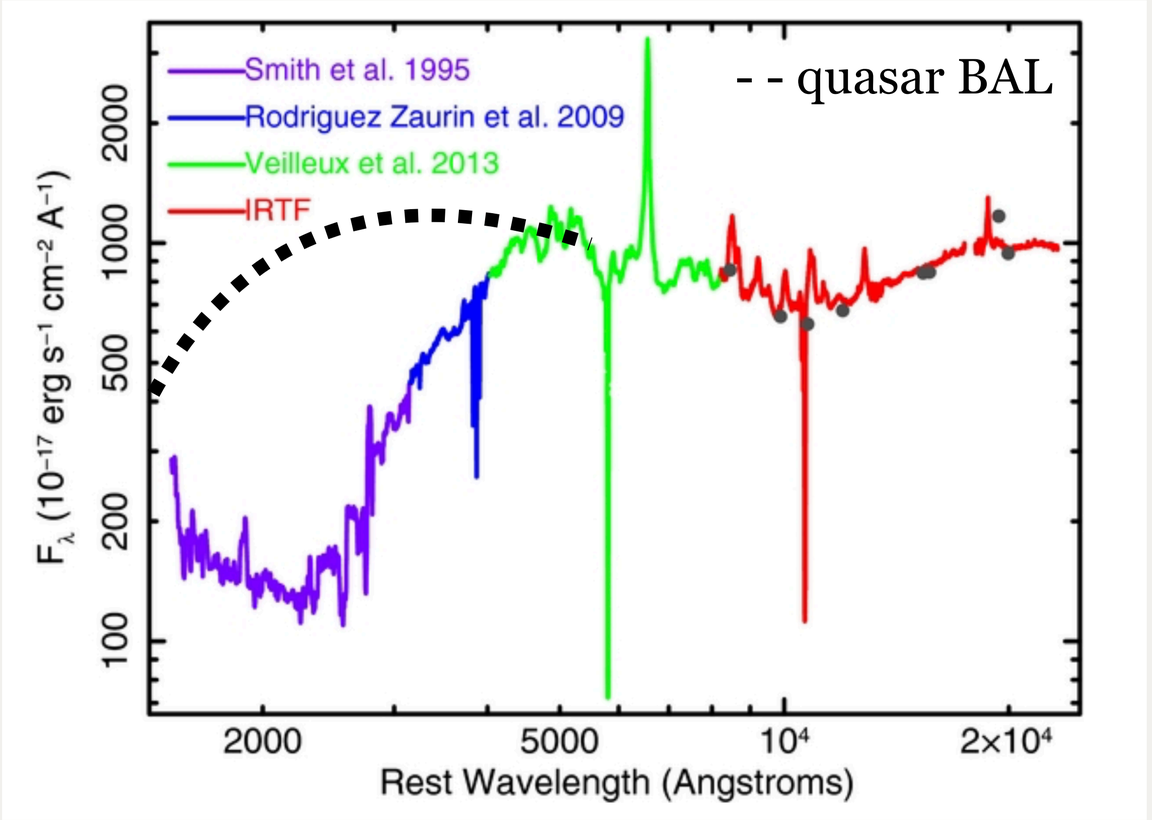


Mrk 231 UV-IR spectrum, *Leighly et al. (2017)*

Introduction - Markarian 231 (Mrk 231)

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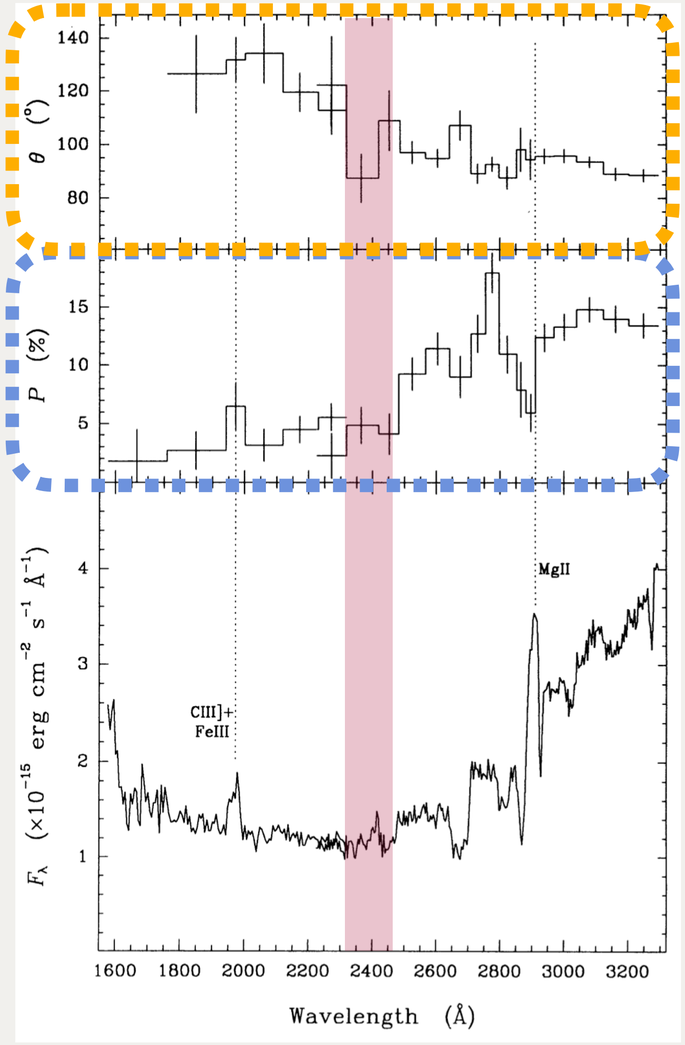
Mrk 231 UV-IR spectrum, *Leighly et al. (2017)*

Introduction - Markarian 231 (Mrk 231)

Mrk 231 - unique polarimetry

- **Ultra luminous infrared galaxy**
  - ➡ z = 0.042 (~ 175 Mpc)
- **Unique UV-optical spectrum**
  - ➡ Cut off ~ 2400 Å
- **Unique UV- optical polarization properties**
  - ➡ Wavelength-dependent polarization degree
  - ➡ Rotation of the polarization angle

➡ **Two sources ?**



Spectropolarimetric data of Mrk 231, HST/FOS, *Smith et al. (1995)*

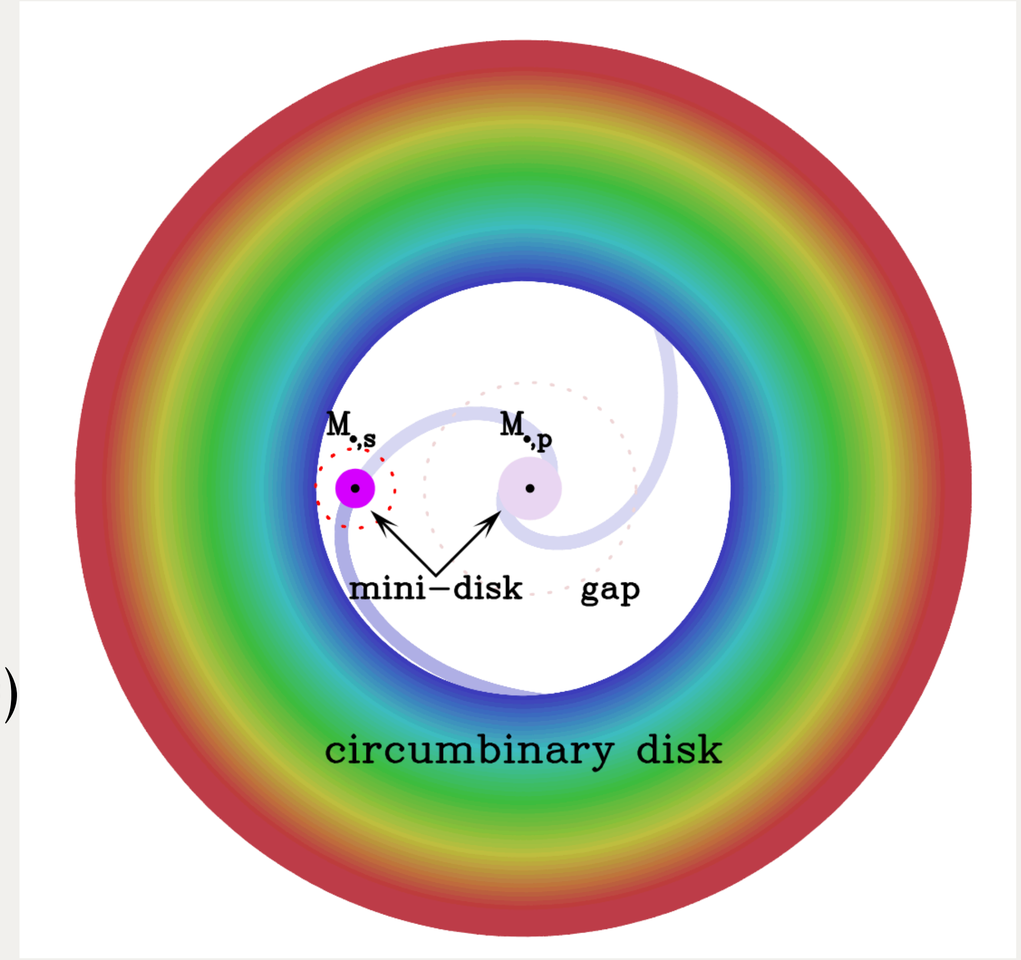
Introduction - Markarian 231 (Mrk 231)

Mrk 231 - Hypothesis for a supermassive black holes binary (SMBH) - Yan and al (2015)

- Binary SMBH in the core of Mrk 231
  - ➡ To explain the cut off in the UV spectrum

- **GOAL**
  - ➡ Reproduce the Spectral Energy Distribution (SED)
  - ➡ Analyse the response of the model in polarization

➡ **Never studied !**

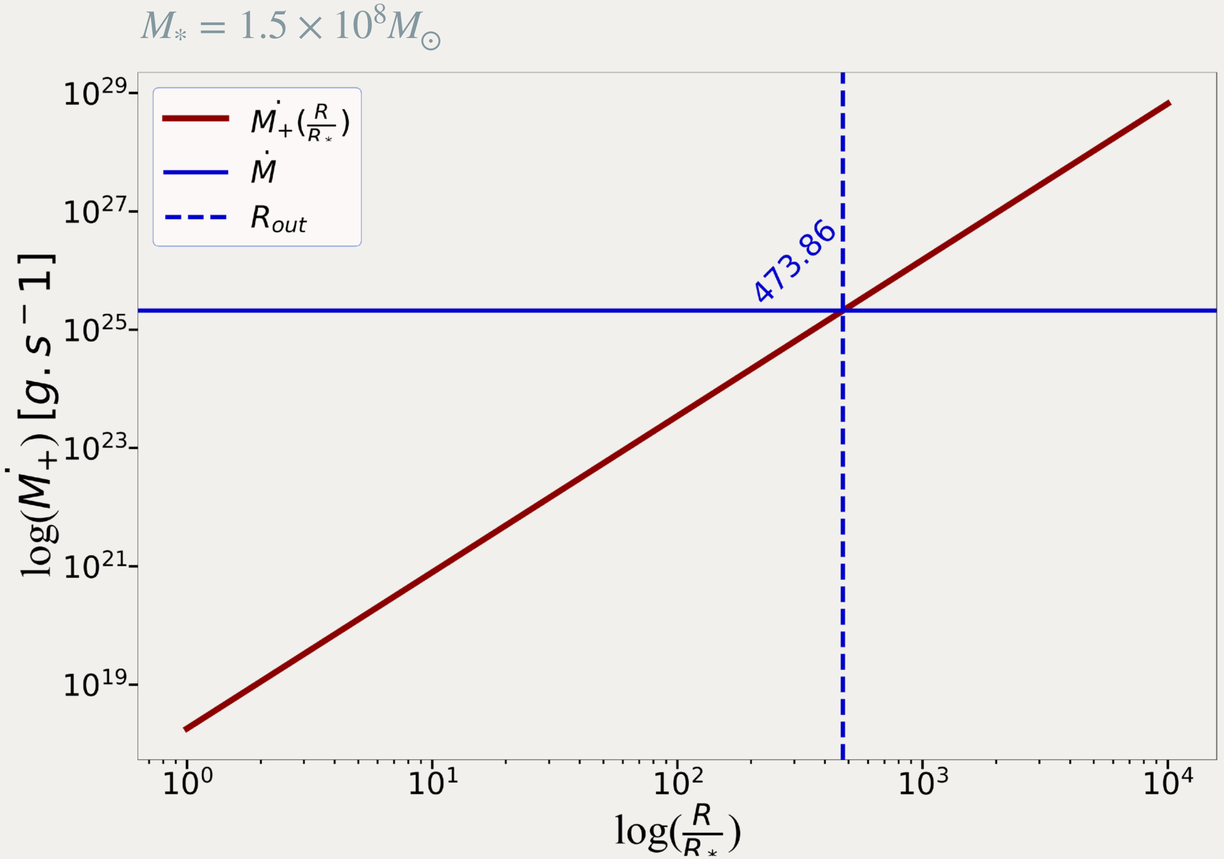


Accretion disk diagram of a supermassive black hole binary system, *Yan et al. (2015)*

Modeling Spectral Energy distribution - Simple SMBH

SED for one SMBH

- Geometrically thin and optically thick disk model
- Multi temperature blackbody radiation
- Outer radius - critical accretion rate $\dot{M}$
  Disk becomes unstable : $\dot{M} > \dot{M}_{+}$ *Lacosta (2016)*



LOG-LOG critical accretion rate as a function of the normalized radius
Biedermann et al. (2025)

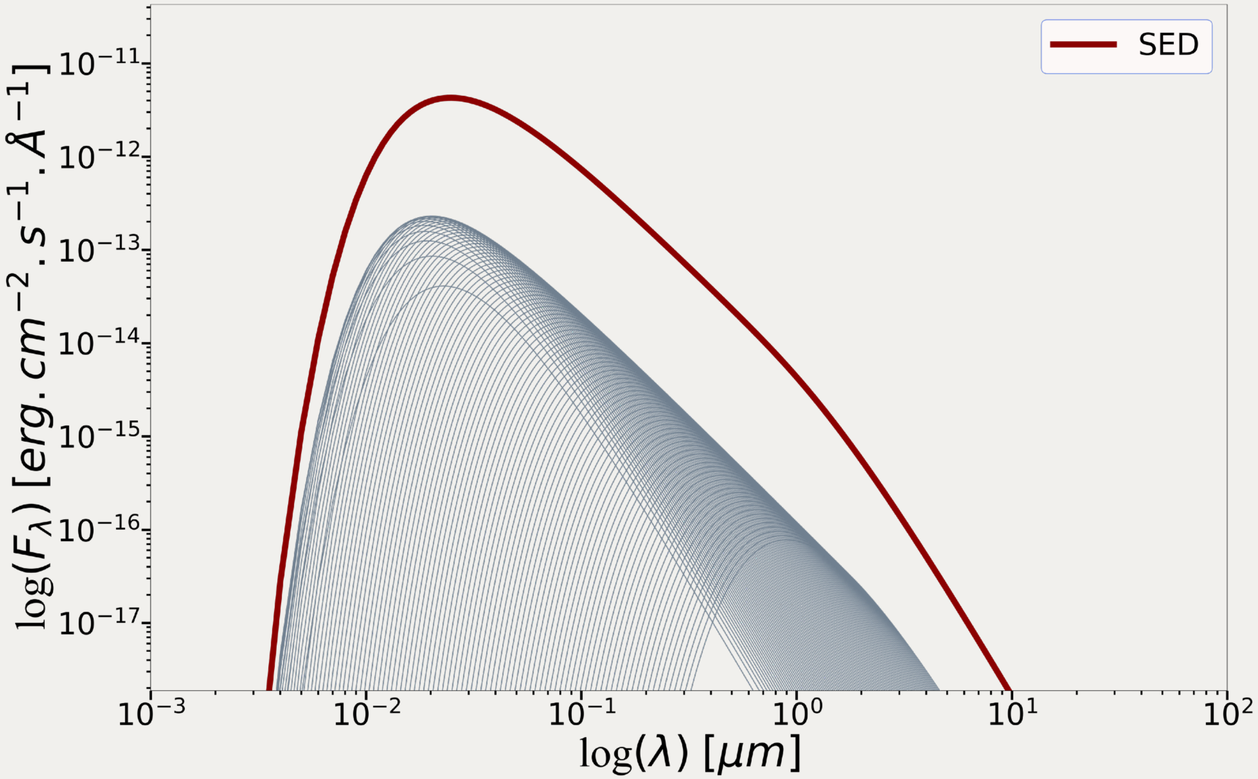
Modeling Spectral Energy distribution - Simple SMBH

SED for one SMBH

- Geometrically thin and optically thick disk model
- Multi temperature blackbody radiation
- Outer radius - critical accretion rate $\dot{M}$
  Disk becomes unstable : $\dot{M} > \dot{M}_{+}$ *Lacosta (2016)*
- SED for a simple SMBH

$$\boldsymbol{\theta} \equiv \begin{pmatrix} M_* \\ \alpha \\ p \\ \theta \end{pmatrix}$$

- M_* ← SMBH mass
- α ← Viscosity
- p ← Accretion parameter
- θ ← Inclination angle



LOG-LOG Flux a function of the wavelength $M_* = 1.5 \times 10^8 M_\odot$
Biedermann et al. (2025)

Modeling Spectral Energy distribution - Binary SMBH

SED for two SMBHs

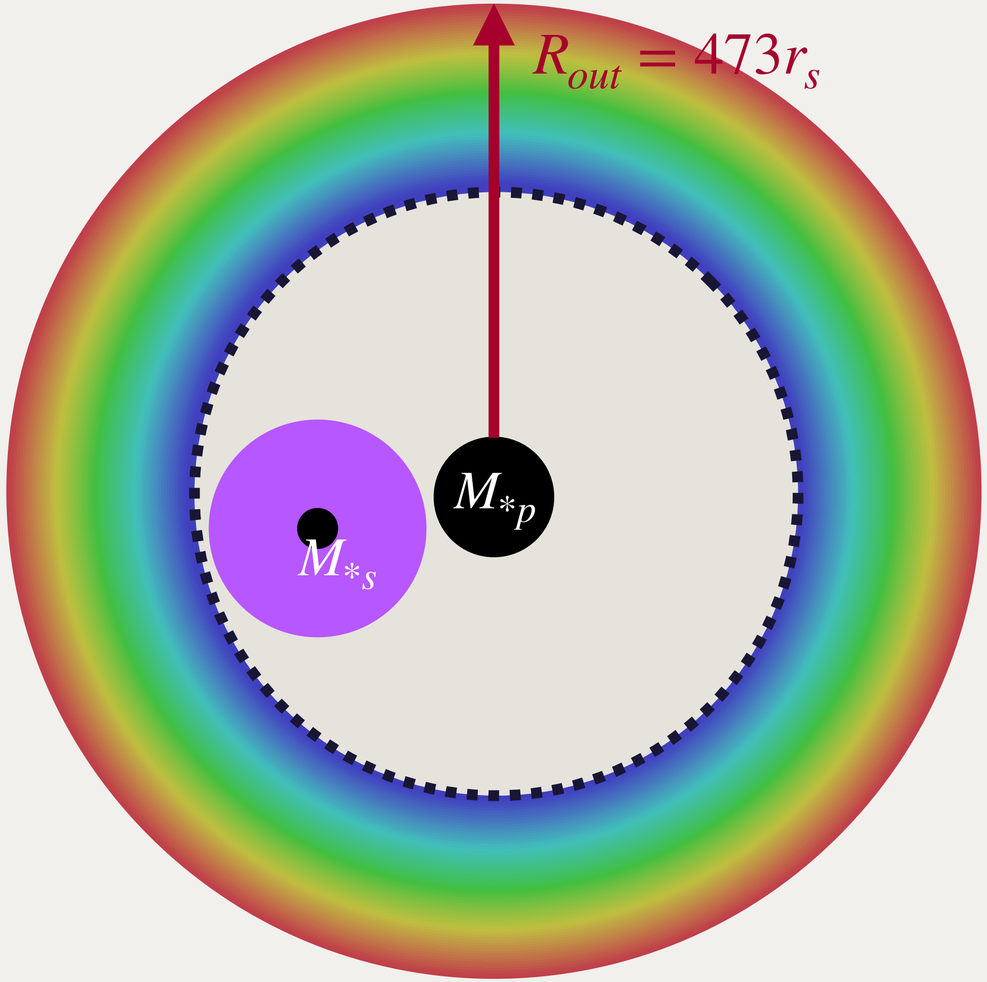
➡ Let's add a secondary SMBH

Primary larger SMBH

- Outer radius - critical accretion rate $\dot{M}_+$
- Inner radius - Hill radius

Secondary smaller SMBH

- Outer radius - Roche lobe radius
- Inner radius - ISCO radius



Accretion disk diagram of a supermassive black hole binary system, inspired from *Yan et al. (2015)*

Modeling Spectral Energy distribution - Binary SMBH

SED for two SMBHs

➡ Let's add a secondary SMBH

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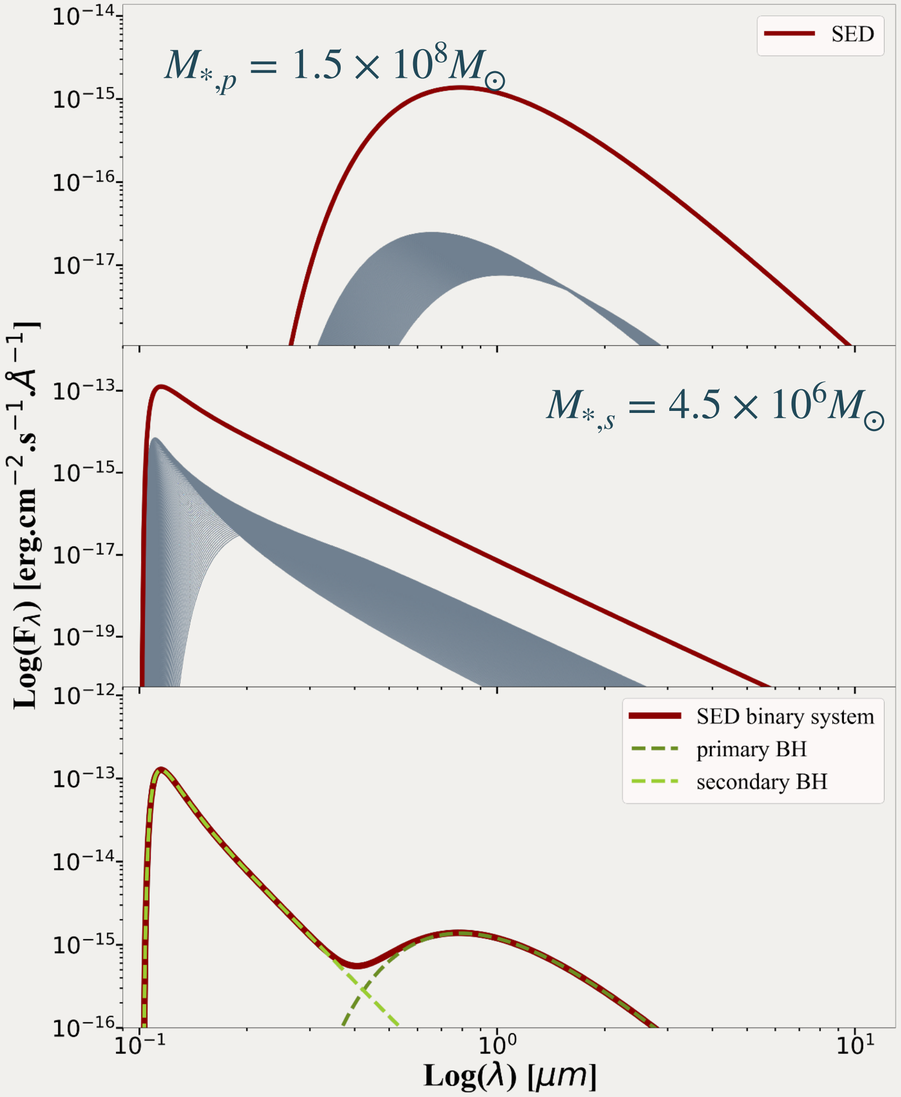
➡ SED for a binary SMBH

Roche lobe fraction

Semi major axis

Primary SMBH Secondary SMBH

$$\boldsymbol{\theta} \equiv \{M_{*,p}, \alpha_p, p_p, \theta_p \; ; \; M_{*,s}, \alpha_s, p_s, \theta_s \; ; \; f_{r,s}, a_{BBH}\}$$



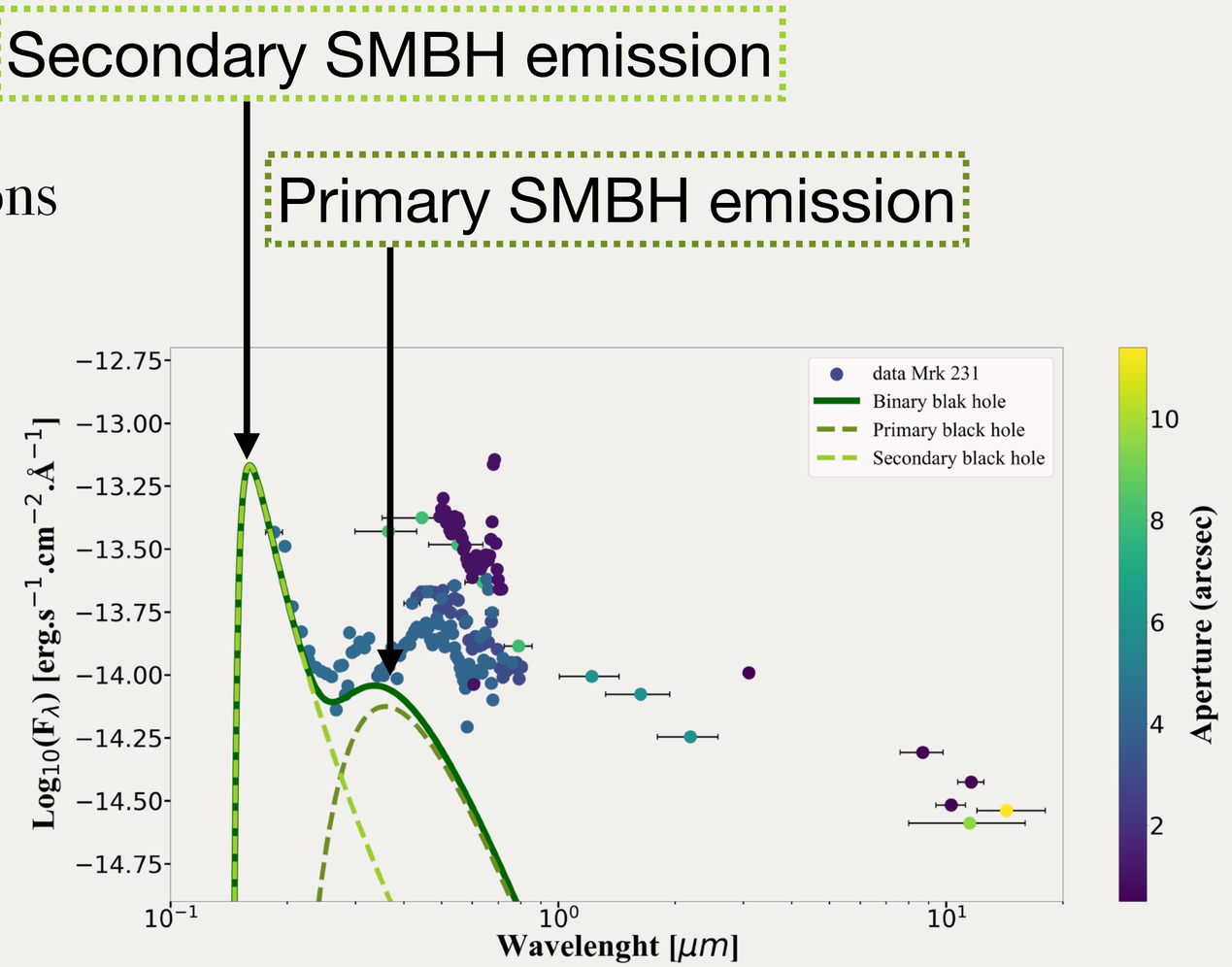
LOG-LOG Flux a function of the wavelength for a binary SED
Biedermann et al. (2025)

Modeling Spectral Energy distribution - Total SED of the quasar

Total SED

- Archival data
  - Flux with corresponding polarimetric observations

- Reproduce the observations
  - Binary SED

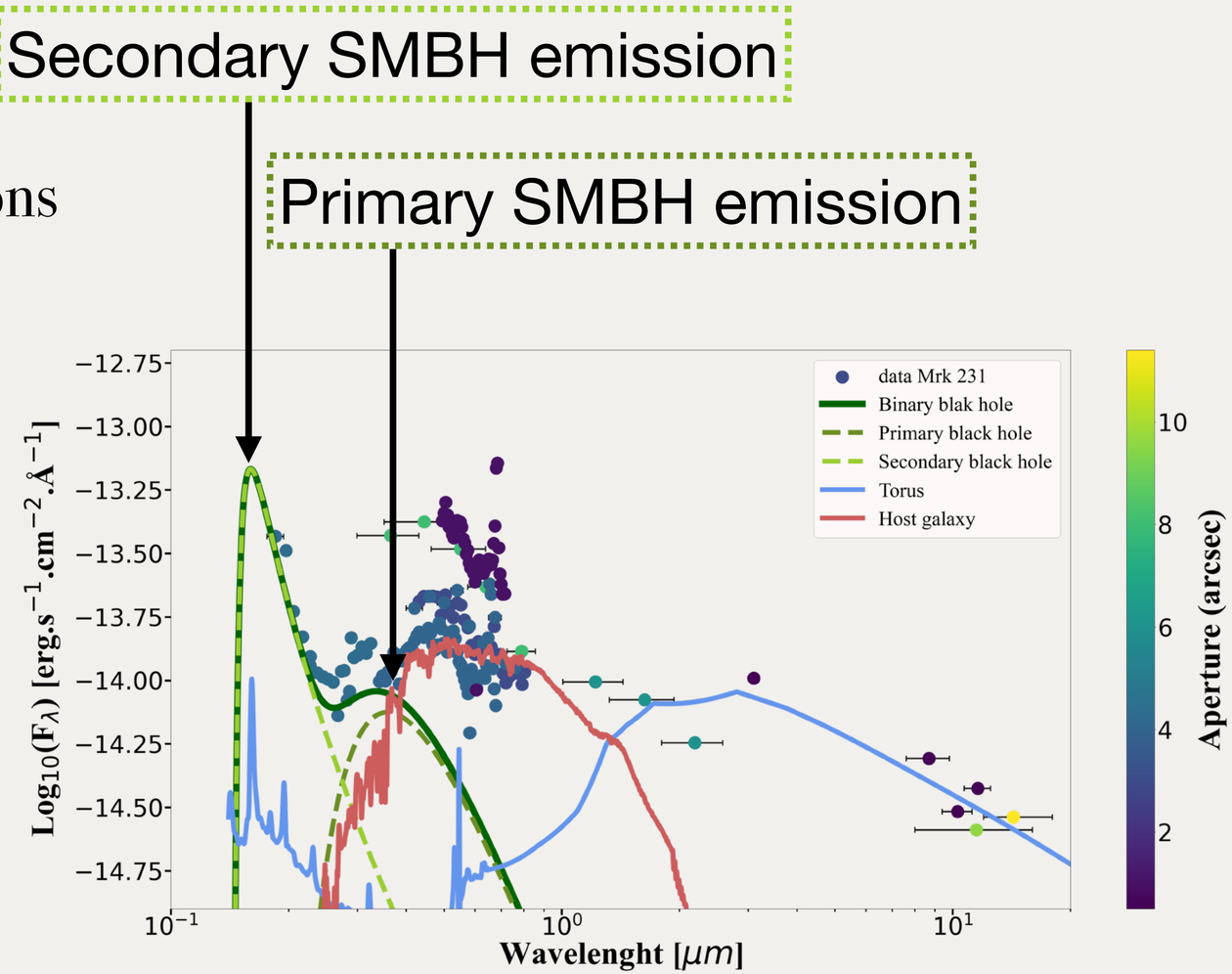


Modeling Spectral Energy distribution - Total SED of the quasar

Total SED

- Archival data
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- Reproduce the observations
  - Binary SED
  - Torus
  - Host galaxy



Modeling Spectral Energy distribution - Total SED of the quasar

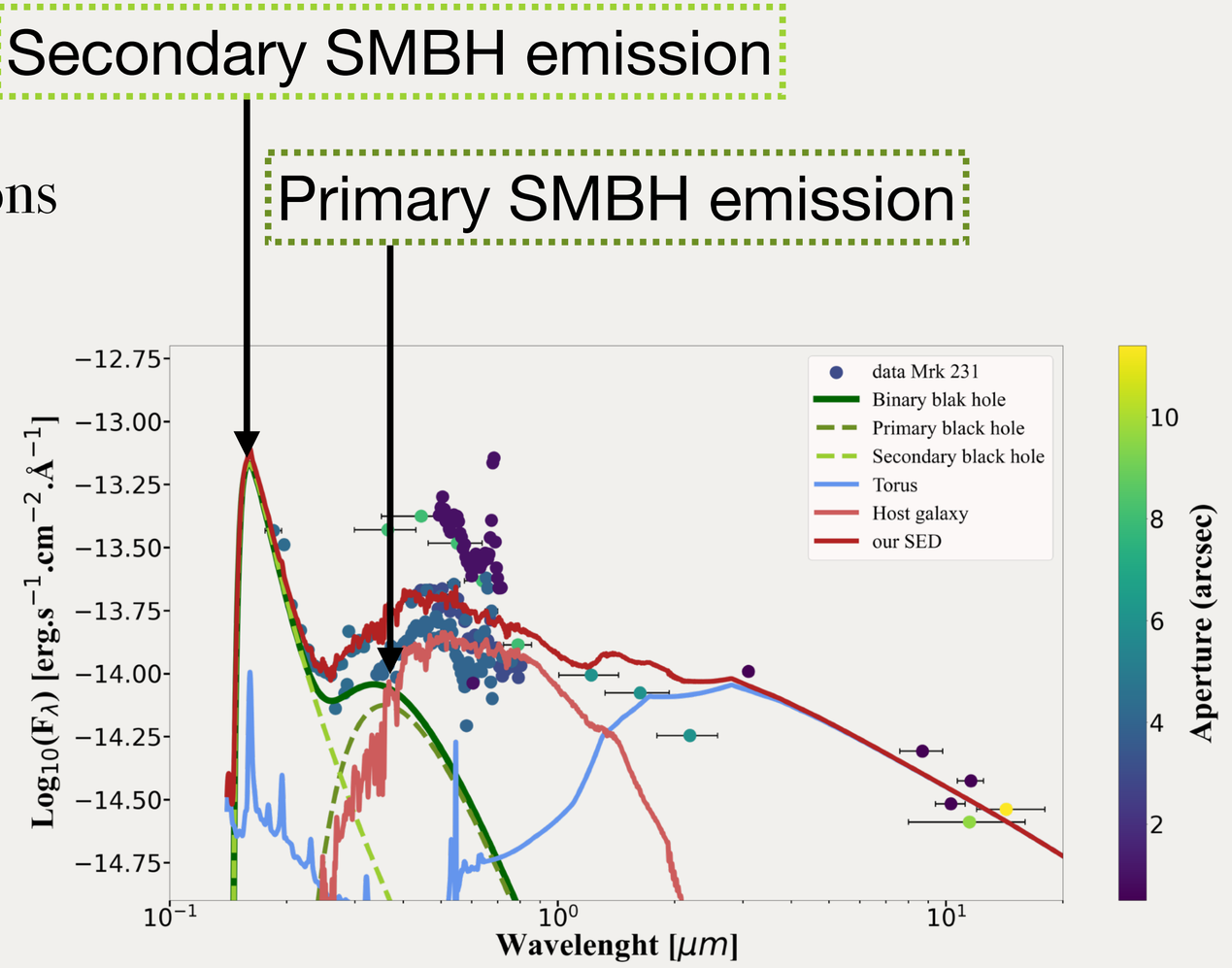
Total SED

➡ Archival data
- Flux with corresponding polarimetric observations

➡ Reproduce the observations
- Binary SED
- Torus
- Host galaxy

➡ Best fit total Spectral Energy Distribution (SED)

➡ **A step further : what is the response of the model in polarization ?**



Polarimetric analysis

- Orientation of the electric field
- Stokes formalism

$$\vec{S} = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

I : Total intensity
Q : Linear polarization $0° \mid 90°$
U : Linear polarization $\pm 45°$
V : Circular polarization

- Polarization adds 2 additional informations to the intensity

  - Polarization degree $P\,[\%] = \dfrac{\sqrt{Q^2 + U^2 + V^2}}{I}$
  - Polarization angle $\theta = \dfrac{1}{2} arctan\left(\dfrac{U}{Q}\right)$

Polarimetry

Polarimetric analysis

➡ For each component of the SED, a best fit polarization value is assigned and the Stokes fluxes are summed.

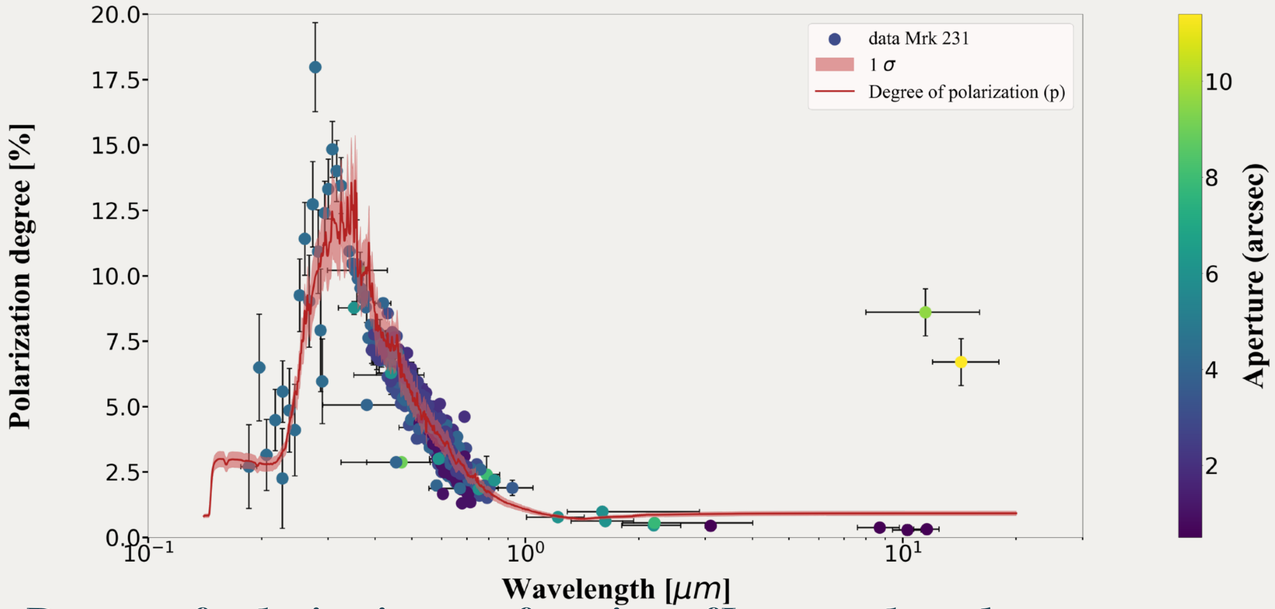
Polarimetry - Degree of polarization

Polarimetric analysis

➡ For each component of the SED, a best fit polarization value is assigned and the Stokes fluxes are summed.

Polarization degree

$$p = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$$



Degree of polarization as a function of Log wavelength
Biedermann et al. (2025)

	Primary Black hole	Secondary Black hole	Host galaxy	Torus
P (%)	23.0 ± 2.7	3.1 ± 0.3	0	0.9 ± 0.1

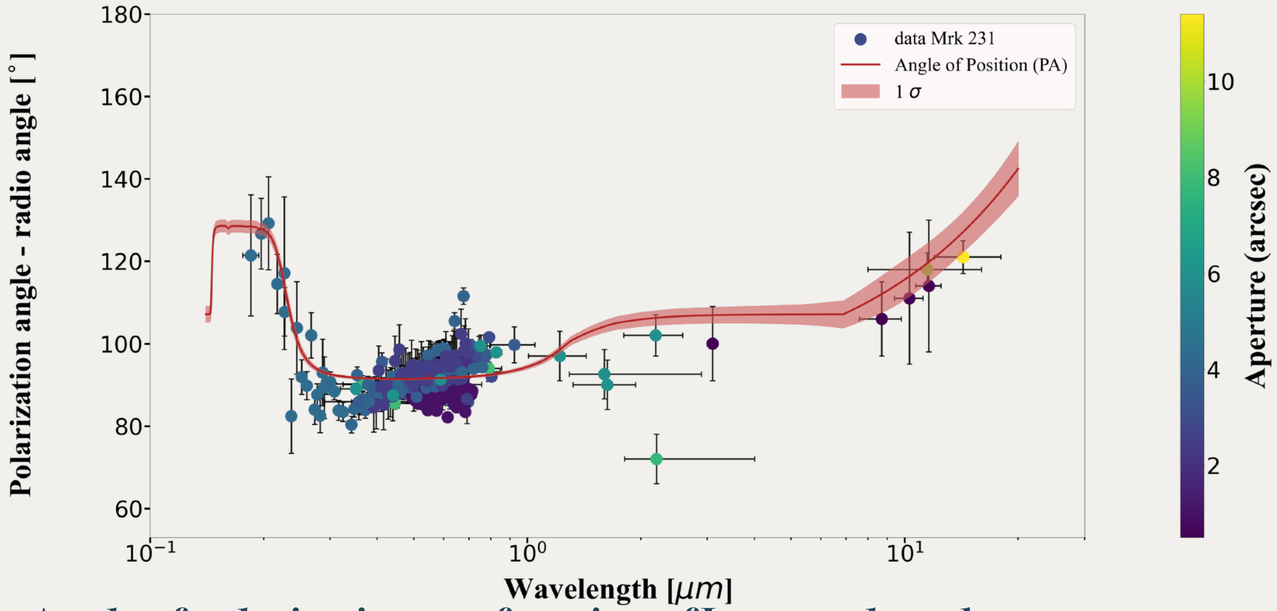
Polarimetry - Angle of polarization

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Polarization degree $p = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$

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Angle of polarization as a function of Log wavelength
Biedermann et al. (2025)

	Primary Black hole	Secondary Black hole	Host galaxy	Torus
P (%)	23.0 ± 2.7	3.1 ± 0.3	0	0.9 ± 0.1
PA (°)	95.9 ± 0.2	133.8 ± 1.5	0	112.1 ± 2.1

Polarimetry - Polarized flux

Polarimetric analysis

➡ For each component of the SED, a best fit polarization value is assigned and the Stokes fluxes are summed.

Polarization degree

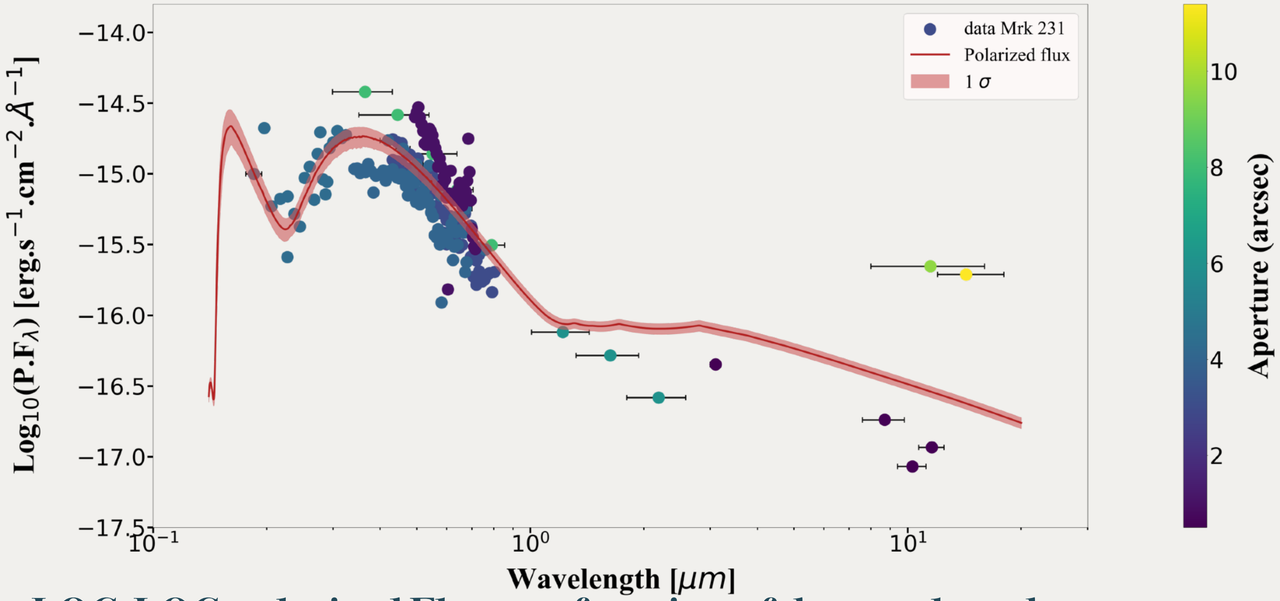
$$p = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}$$

Polarization angle

$$\theta = \frac{1}{2} arctan\left(\frac{U}{Q}\right)$$

Polarized flux

➡ Strong prediction for a second bump in FUV



LOG-LOG polarized Flux as a function of the wavelength
Biedermann et al. (2025)

	Primary Black hole	Secondary Black hole	Host galaxy	Torus
P (%)	23.0 ± 2.7	3.1 ± 0.3	0	0.9 ± 0.1
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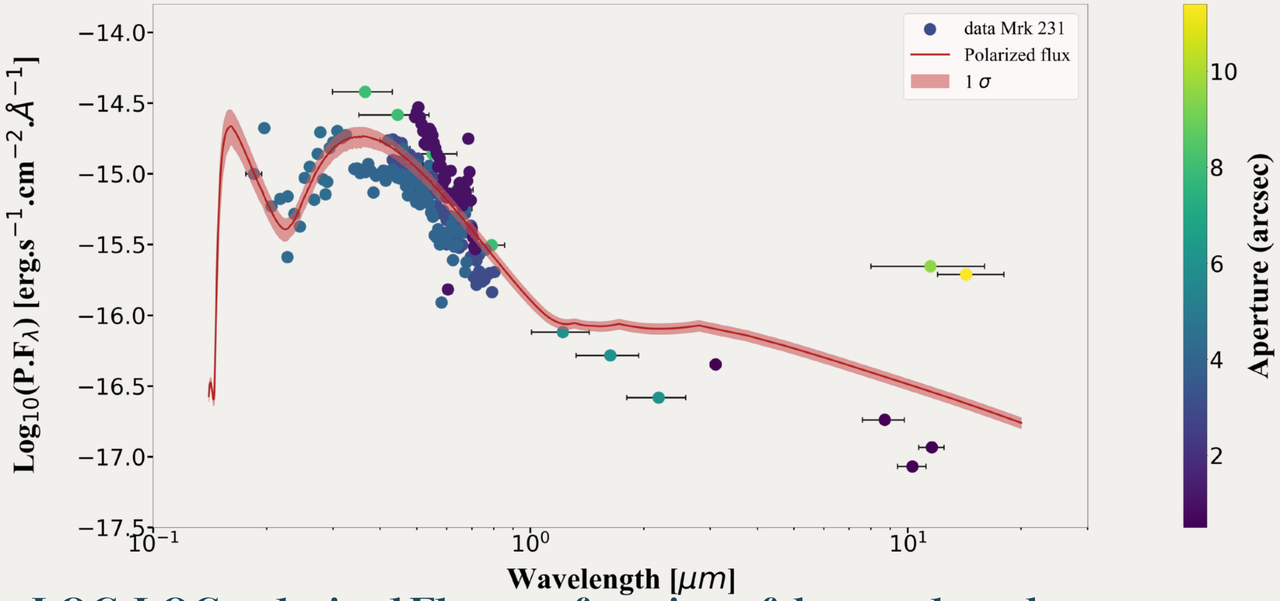
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Polarized flux

➡ Strong prediction for a second bump in FUV

Model reproduces three independent measurements



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Summary - Probing binary SMBH in quasars through spectropolarimetry

➡ Recreate the total SED of Mrk 231 using a simple binary SMBH model

➡ Binary SMBH model in agreement with our spectropolarimetric data

➡ Next step to test the binary hypothesis

✓ To probe the variation in time of Lyα profile

✓ To confirm/reject the presence of a second bump in FUV most easily detectable in polarized light

($\lambda_{rest} \sim 1460 - 2350$ Å)

➡ Future high-sensitivity UV spectropolarimeters such as the instrument like POLLUX proposed for the Habitable Worlds Observatory

THANK YOU

FOR YOUR ATTENTION

Questions or feedback ?

BACK SLIDE

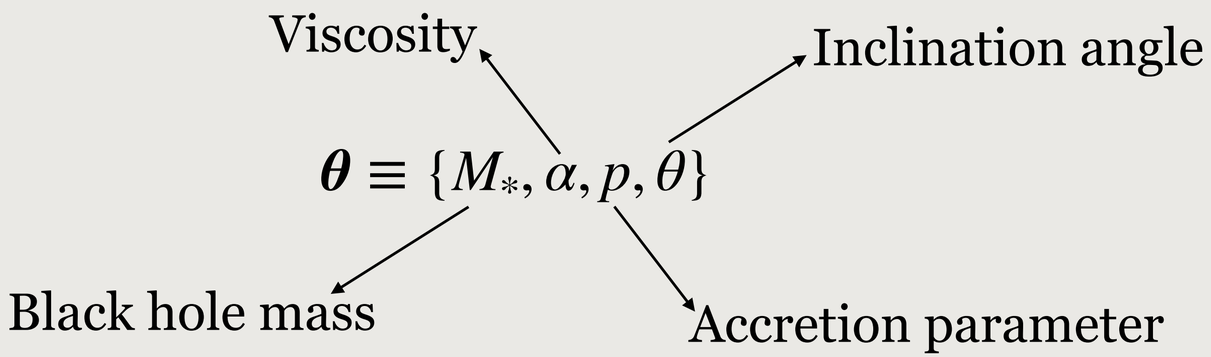
- Theoretical radius - Toomre parameter indicates a local gravitational instability in the disc
  - Disk become unstable : $\dot{M} > \dot{M}_{+}$ *Lacosta (2016)*

$$\dot{M}_{+} = 8.07 \times 10^{15} \alpha_{0.1}^{-0.01} \left(\frac{r}{10^{10}[cm]} \right)^{2.64} \left(\frac{M_*}{M_\odot} \right)^{-0.89}$$

$$\dot{M} = \alpha p \dot{M}_{Eddington}$$

- Black hole masse : M_*
- Viscosity : α
- Accretion parameter : p
- Radius of the accretion disk : r
- Eddington accretion rate : $\dot{M}_{Eddington}$

- $\dot{M}$ is linked to thermal instabilities



BACK SLIDE

❖Inner radius for the primary SMBH

$$R_{int,p} = \frac{a_{BBH}}{1+q} + a_{BBH}\left(\frac{M_{*,s}}{3M_{*,p}}\right)^{\frac{1}{3}}$$

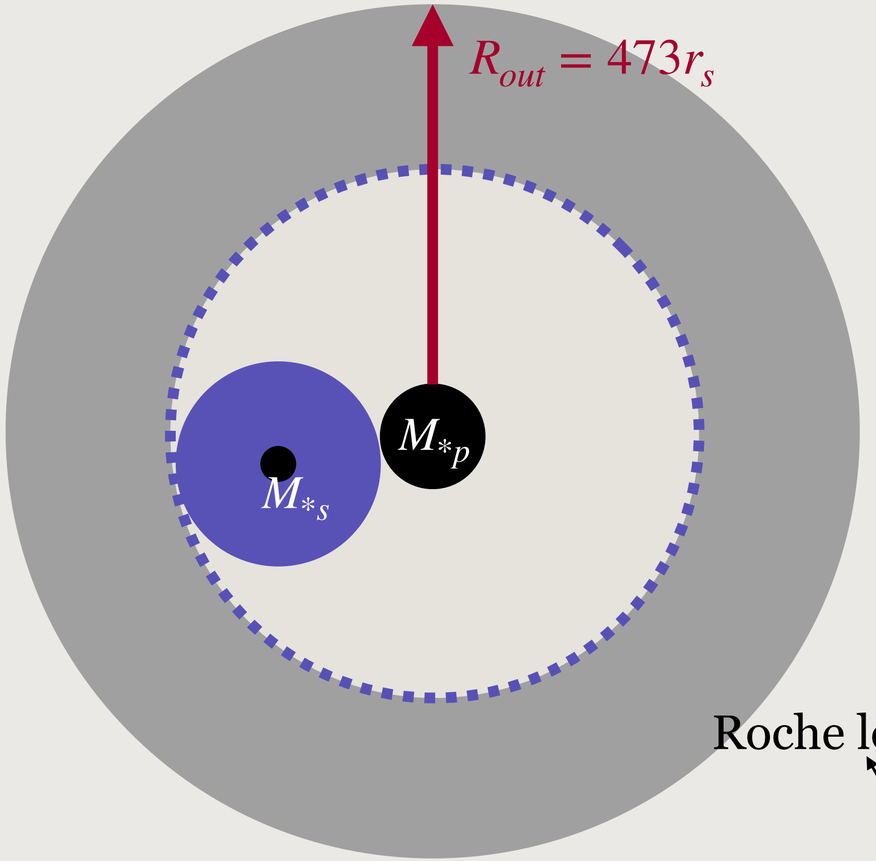
Yan et al (2015)

❖Outer radius for the secondary SMBH

$$R_{out,s} = f_{r,s} \times \left(\frac{0.49 a_{BBH} q^{\frac{2}{3}}}{0.6 q^{\frac{2}{3}} + log(1+q^{\frac{1}{3}})}\right)$$

Yan et al (2015)

$$R_{int,s} = r_s$$



$$\boldsymbol{\theta} \equiv \{\underbrace{M_{*,p}, \alpha_p, p_p, \theta_p}_{\text{Primary SMBH}} ; \underbrace{M_{*,s}, \alpha_s, p_s, \theta_s}_{\text{Secondary SMBH}} ; f_{r,s}, a_{BBH}\}$$

Roche lobe fraction

Semi major axis